
MIDTERM EXAM 3 (FORM B)

NAME: SOLUTIONS

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- When solving a problem, if you are going to use an alternative method to the ones discussed in class (for instance, inverting a matrix via adjugates or using Cramer's rule to solve a linear system), you are required to include a short explanation on how this alternative method works.
- The use of calculators of any type is not allowed.

This exam consists of four problems, and your total score will be out of $10 + 10 + 10 + 10 = 40$ points.

Good Luck!

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Problem 1 (10 points). Answer the following questions concerning linear independence.

- (a) (5 points) Let V be a vector space and suppose $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a set of vectors in V . State precisely what it means for the set $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ to be a linearly independent set in V .

The set $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k\}$ is linearly independent if whenever we have a linear combination

$$a_1 \vec{x}_1 + a_2 \vec{x}_2 + \dots + a_k \vec{x}_k = \vec{0}, \text{ where } a_1, a_2, \dots, a_k \in \mathbb{R},$$

then $a_1 = a_2 = \dots = a_k = 0$.

- (b) (5 points) Suppose $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is a linearly independent set in V . Determine if $\{\mathbf{x}, \mathbf{y} - \mathbf{x}, \mathbf{z}\}$ is a linearly independent set.

Is $\{\vec{x}, \vec{y} - \vec{x}, \vec{z}\}$ linearly independent? Suppose $a\vec{x} + b(\vec{y} - \vec{x}) + c\vec{z} = \vec{0}$ for some scalars $a, b, c \in \mathbb{R}$. Then:

$$a\vec{x} + b(\vec{y} - \vec{x}) + c\vec{z} = \vec{0} \Rightarrow a\vec{x} + b\vec{y} - b\vec{x} + c\vec{z} = \vec{0} \Rightarrow (a-b)\vec{x} + b\vec{y} + c\vec{z} = \vec{0} \Rightarrow \begin{matrix} \uparrow \\ \{\vec{x}, \vec{y}, \vec{z}\} \text{ independent} \end{matrix} \quad a-b=0, b=0, c=0$$

This implies $a=b=0, c=0$, so we conclude:

$\{\vec{x}, \vec{y} - \vec{x}, \vec{z}\}$ is linearly independent.

Problem 2 (10 points). Let $\mathbf{P}_3$ be the vector space of all polynomials of degree at most 3, including the zero polynomial. We define a transformation $T: \mathbf{P}_3 \rightarrow \mathbb{R}^2$ via the equation

$$T(p(x)) = (p(0), p(2)), \quad \text{for all } p(x) \in \mathbf{P}_3.$$

(a) (4 points) Show that T is a linear transformation. (The question continues on the next page)

We have to check two properties:

1) Is $T((p+q)(x)) = T(p(x)) + T(q(x))$ for all $p(x) \in \mathbf{P}_3$?

Let $p(x), q(x) \in \mathbf{P}_3$ be arbitrary. Then:

$$\begin{aligned} T((p+q)(x)) &= ((p+q)(0), (p+q)(2)) = (p(0)+q(0), p(2)+q(2)) = (p(0), p(2)) + (q(0), q(2)) \\ &= T(p(x)) + T(q(x)) \quad \checkmark \end{aligned}$$

2) Is $T((ap)(x)) = aT(p(x))$ for all $p(x) \in \mathbf{P}_3$ and $a \in \mathbb{R}$?

$$T((ap)(x)) = ((ap)(0), (ap)(2)) = (ap(0), ap(2)) = a(p(0), p(2)) = aT(p(x)) \quad \checkmark$$

Thus, T is linear because it respects addition and scalar multiplication.

(b) (6 points) Compute a basis and the dimension of the kernel $\ker T$.

We first need to describe the elements of $\ker T = \{p(x) \in P_3 \mid T(p(x)) = 0\}$

Suppose $p(x) = a + bx + cx^2 + dx^3$ is in P_3 . Then:

$$p(x) \in \ker T \Leftrightarrow T(p(x)) = (p(0), p(2)) = (0, 0) \Leftrightarrow (a, a + 2b + 4c + 8d) = (0, 0)$$

$$\Leftrightarrow \begin{cases} a = 0 \\ a + 2b + 4c + 8d = 0 \end{cases} \quad \text{The solutions to this system are} \quad \begin{cases} a = 0 \\ b = -2s - 4t \\ c = s \\ d = t \end{cases}, s, t \in \mathbb{R}$$

$$\text{So } p(x) \in \ker T \Leftrightarrow p(x) = (-2s - 4t)x + sx^2 + tx^3 = s(x^2 - 2x) + t(x^3 - 4x), \quad s, t \in \mathbb{R}.$$

$\Rightarrow \{x^2 - 2x, x^3 - 4x\}$ is a spanning set for $\ker T$. Is the set independent?

$$a(x^2 - 2x) + b(x^3 - 4x) = 0 \Rightarrow (-2a - 4b)x + ax^2 + bx^3 = 0 \Rightarrow a = b = 0$$

$a, b \in \mathbb{R}$ $\nwarrow \quad \uparrow \quad \nearrow$
Coefficients

Thus, the set is independent. We conclude:

- Basis of $\ker T$: $\{x^2 - 2x, x^3 - 4x\}$
- $\dim(\ker T) = 2$

Problem 3 (10 points). Consider the matrix

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

The characteristic polynomial of A is $c_A(x) = (x-1)(x-3)^2$.

- (a) (2 points) The matrix A has two eigenvalues λ_1 and λ_2 with $\lambda_1 > \lambda_2$. What are their values?

The eigenvalues of A are the solutions to $c_A(x) = (x-1)(x-3)^2 = 0$

$$\lambda_1 = 3, \quad \lambda_2 = 1$$

- (b) (4 points) Find a set of basic eigenvectors for A associated with the eigenvalue λ_1 .
(The question continues on the next page)

The eigenvector associated with $\lambda_1 = 3$ are the solutions $\vec{x}$ to $(3I - A)\vec{x} = \vec{0}$.

$$3I - A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \Rightarrow \text{The } 3\text{-eigenvectors are the vectors } \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ with } \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{The system is } \begin{cases} y - z = 0 \\ y - z = 0 \end{cases} \Rightarrow \begin{cases} x = s \\ y = z = t, \quad t \in \mathbb{R} \\ z = t \end{cases}$$

So the 3-eigenvectors are

$$\begin{bmatrix} s \\ t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{R}, \quad s \neq 0 \text{ or } t \neq 0.$$

Thus, a set of basic eigenvectors for A with eigenvalue 3 is

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

- (c) (4 points) You are given that the vector $\mathbf{z} = (0, 1, -1)$ is an eigenvector of A with eigenvalue λ_2 . Is A diagonalizable? If so, find a diagonal matrix D and an invertible matrix P such that $D = P^{-1}AP$. Otherwise, explain why A is not diagonalizable.

The matrix $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ is invertible ($\det P = -2 \neq 0$) and it consists of eigenvectors of A , so $P^{-1}AP$ is the diagonal matrix $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (The diagonal entries are the eigenvalues of A corresponding to each column)

Therefore:

$$A \text{ is diagonalizable; } P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \text{ and } P^{-1}AP = D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 4 (10 points). Let

$$B = \begin{bmatrix} 1 & 0 & 2 & 0 \\ -2 & 2 & -6 & 14 \\ -3 & 5 & -11 & 35 \end{bmatrix}, \text{ whose reduced row-echelon form is } R = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

- (a) (5 points) Find a basis for the row space $\text{row } B$ and a basis for the column space $\text{col } B$.

A basis of row B is the set of nonzero rows of R :

$$\text{Basis of row } B: \{(1, 0, 2, 0), (0, 1, -1, 7)\}$$

A basis of $\text{col } B$ is the set of columns of B whose corresponding columns of R contain leading 1's:

$$R = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{Basis of col } B: \{(1, -2, -3), (0, 2, 5)\}$$

$\uparrow \uparrow$
 Leading 1's

- (b) (5 points) State precisely what is the null space of B , $\text{null } B$. Then, compute the dimension of $\text{null } B$.

(**Hint:** you are not asked to find $\text{null } B$ explicitly).

$$\text{The null space of } B \text{ is } \text{null } B = \{\vec{x} \in \mathbb{R}^4 \mid B\vec{x} = \vec{0}\}$$

To compute $\dim(\text{null } B)$, we can use the rank-nullity theorem: since B is a 3×4 matrix:

$$\dim(\text{null } B) + \underbrace{\dim(\text{col } B)}_2 = \underbrace{n^\circ \text{ of columns of } B}_4 \Rightarrow \dim(\text{null } B) = 2$$

Scratch working page

Scratch working page